COHOMOLOGICAL DESCENT FOR PROPER HYPERCOVERINGS ON THE ZARISKI SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lemma-compare-simplicial-objects, lemma-direct-image-morphism-simplicial-sites, lemma-hypercovering-X-simple-descent-bounded-abelian; external tools from Cohomology on Sites.

Editorial note. The label lemma-cohomological-descent-for-proper-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Comparison diagram. Same unit, Lemma lemma-compare-simplicial-objects. Commutative diagram relating a and a_qc used for the second equality.
- qc hypercovering descent on components. Cohomology on Sites, Lemma sites-cohomology-lemma-cohomological-descent-LC. Rh_{n,*}h_n^{-1} = id on D^+(U_n); also collapses the last step.
- Simplicial direct image. Same unit, Lemma lemma-direct-image-morphism-simplicial-sites. Rh_*h^{-1} = id on D^+(U_Zar) follows levelwise.
- Descent for qc hypercoverings. Same unit, Lemma lemma-hypercovering-X-simple-descent-bounded-abelian. Third equality: Ra_{qc,*}a_{qc}^{-1} = id (U is a hypercovering in LC_qc by the Cohomology on Sites proper-surjective lemma).

Source reference. Original reference 09W4: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 524–524. Prerequisite pointer invoked at proof line(s) 6050; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DAF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5910–5910. Prerequisite pointer invoked at proof line(s) 6054; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0D97: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1353–1353. Prerequisite pointer invoked at proof line(s) 6060; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D8N: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5086–5086. Prerequisite pointer invoked at proof line(s) 6076; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D91: sources/stacks-chow/tags.txt, source lines 16611–16611. Prerequisite pointer invoked at proof line(s) 6057; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09X5: sources/stacks-chow/tags.txt, source lines 12518–12518. Prerequisite pointer invoked at proof line(s) 6079; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let U be a simplicial object of LC and let $a : U \to X$ be an augmentation. If $a : U \to X$ gives a proper hypercovering of X, then for $K \in D^+(X)$*

$$K \to Ra_*(a^{-1}K)$$

is an isomorphism where $a : Sh(U_{Zar}) \to Sh(X)$ is as in Lemma 09W4.

Proof. Consider the diagram of Lemma 0DAF. Observe that $Rh_{n,*}h_n^{-1}$ is the identity functor on $D^+(U_n)$ by Cohomology on Sites, Lemma 0D91. Hence Rh_*h^{-1} is the identity functor on $D^+(U_{Zar})$ by Lemma 0D97. We have

$$\begin{aligned} Ra_*(a^{-1}K) &= Ra_*Rh_*h^{-1}a^{-1}K \\ &= Rh_{-1,*}Ra_{qc,*}a_{qc}^{-1}(h_{-1})^{-1}K \\ &= Rh_{-1,*}(h_{-1})^{-1}K \\ &= K \end{aligned}$$

The first equality by the discussion above, the second equality because of the commutativity of the diagram in Lemma 0DAF, the third equality by Lemma 0D8N (U is a hypercovering of X in LC_{qc} by Cohomology on Sites, Lemma 09X5), and the last equality by the already used Cohomology on Sites, Lemma 0D91. □